Binet's Formula via Matrix Diagonalization

Linear Algebra Honours — Project Report

Abstract. The Fibonacci sequence $\{F_n\}_{n\geq 0}$, defined by the recurrence $F_{n+2} = F_{n+1} + F_n$ with $F_0 = 0$ and $F_1 = 1$, appears throughout mathematics, nature, and computer science. This report derives a closed-form expression for F_n, known as *Binet's formula*, using the technique of matrix diagonalization. Along the way we resolve a striking paradox: how a formula filled with irrational numbers produces an integer for every $n \in \mathbb{N}_0$. We conclude with a proof by strong induction that the formula is universally valid, two concrete applications, and a resolution of the integrality paradox.

1. Introduction and Motivation

The Fibonacci sequence begins

$$0,\ 1,\ 1,\ 2,\ 3,\ 5,\ 8,\ 13,\ 21,\ 34,\ \ldots$$

and is completely determined by the recurrence relation

$$F_{n+2} = F_{n+1} + F_n, \qquad F_0 = 0, \quad F_1 = 1.$$

Computing F_n iteratively is straightforward but slow for large n: each new term requires all previous terms. A natural question is therefore whether a *closed-form* expression exists—one that computes any term directly, without stepping through all previous values.

The answer is yes, and the key tool is **matrix diagonalization**. The central result of this report is the following.

Theorem 1.1 (Binet's Formula)**.** *For every integer $n \geq 0$,*

$$F_n \;=\; \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right).$$

Sections 2–3 develop and verify this theorem in full. Section 4 explains why the formula always yields an integer despite containing $\sqrt{5}$ throughout.

2. Proof of Binet's Formula

Throughout the proof we use the shorthand

$$\varphi \;\coloneqq\; \frac{1+\sqrt{5}}{2} \qquad \text{(the golden ratio)}, \qquad \psi \;\coloneqq\; \frac{1-\sqrt{5}}{2}.$$

2.1. Step 1 — The Matrix Recurrence

Definition 2.1. The *state vector* at step n is $\mathbf{v}_n := \begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix}$, with $\mathbf{v}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Lemma 2.2. *Let* $M := \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. *Then* $\mathbf{v}_{n+1} = M\,\mathbf{v}_n$ *for all* $n \geq 0$*, and consequently* $\mathbf{v}_n = M^n \mathbf{v}_0$.

Proof. Direct computation gives

$$M\,\mathbf{v}_n \;=\; \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} \;=\; \begin{pmatrix} F_{n+1} + F_n \\ F_{n+1} \end{pmatrix} \;=\; \begin{pmatrix} F_{n+2} \\ F_{n+1} \end{pmatrix} \;=\; \mathbf{v}_{n+1},$$

where the third equality uses the recurrence $F_{n+2} = F_{n+1} + F_n$. The formula $\mathbf{v}_n = M^n \mathbf{v}_0$ then follows by induction on n. □

2.2. Step 2 — Eigenvalues of M

We compute the characteristic polynomial of M:

$$\det(M - \lambda I) \;=\; \det \begin{pmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{pmatrix} \;=\; (1-\lambda)(-\lambda) - 1 \;=\; \lambda^2 - \lambda - 1.$$

Applying the quadratic formula to $\lambda^2 - \lambda - 1 = 0$ yields two distinct real eigenvalues:

$$\lambda_1 = \varphi = \frac{1+\sqrt{5}}{2}, \qquad \lambda_2 = \psi = \frac{1-\sqrt{5}}{2}.$$

Remark 2.3. By Vieta's formulas for $\lambda^2 - \lambda - 1 = 0$:

$$\varphi + \psi = 1, \qquad \varphi\psi = -1, \qquad \varphi - \psi = \sqrt{5}.$$

Both roots satisfy the identity $x^2 = x + 1$, which is used repeatedly below.

2.3. Step 3 — Eigenvectors

Eigenvector for φ. We solve $(M - \varphi I)\mathbf{x} = \mathbf{0}$. Row-reducing:

$$M - \varphi I = \begin{pmatrix} 1-\varphi & 1 \\ 1 & -\varphi \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & -\varphi \\ 1-\varphi & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - (1-\varphi)R_1} \begin{pmatrix} 1 & -\varphi \\ 0 & 0 \end{pmatrix}.$$

The bottom-right entry after elimination is $1 + \varphi - \varphi^2$. Since $\varphi^2 = \varphi + 1$ (the characteristic equation), this equals $1 + \varphi - (\varphi + 1) = 0$, confirming the zero row. From $x_1 = \varphi\, x_2$ we choose $x_2 = 1$, giving

$$\mathbf{x}_1 = \begin{pmatrix} \varphi \\ 1 \end{pmatrix}.$$

Eigenvector for ψ. The identical argument (using $\psi^2 = \psi + 1$) yields

$$\mathbf{x}_2 = \begin{pmatrix} \psi \\ 1 \end{pmatrix}.$$

2.4. Step 4 — Diagonalizing Matrix P and its Inverse

Form the modal matrix P by placing the eigenvectors as columns:

$$P = \begin{pmatrix} \varphi & \psi \\ 1 & 1 \end{pmatrix}.$$

Its determinant is $\det(P) = \varphi - \psi = \sqrt{5}$. By the $2{\times}2$ inverse formula,

$$P^{-1} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -\psi \\ -1 & \varphi \end{pmatrix}.$$

One verifies that $P^{-1}MP = D$ where $D = \operatorname{diag}(\varphi,\, \psi)$.

2.5. Step 5 — Computing M^n and Extracting the Formula

Since $M = PDP^{-1}$, raising both sides to the n-th power gives

$$M^n = PD^nP^{-1}, \qquad D^n = \begin{pmatrix} \varphi^n & 0 \\ 0 & \psi^n \end{pmatrix}.$$

Applying this to the initial vector $\mathbf{v}_0$:

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = \mathbf{v}_n = PD^nP^{-1}\,\mathbf{v}_0.$$

We compute the product from right to left. First,

$$P^{-1}\mathbf{v}_0 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -\psi \\ -1 & \varphi \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Then

$$\begin{aligned} \mathbf{v}_n &= \frac{1}{\sqrt{5}}\, PD^n \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &= \frac{1}{\sqrt{5}} \begin{pmatrix} \varphi & \psi \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \varphi^n & 0 \\ 0 & \psi^n \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &= \frac{1}{\sqrt{5}} \begin{pmatrix} \varphi & \psi \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \varphi^n \\ -\psi^n \end{pmatrix} \\ &= \frac{1}{\sqrt{5}} \begin{pmatrix} \varphi^{n+1} - \psi^{n+1} \\ \varphi^n - \psi^n \end{pmatrix}. \end{aligned}$$

Reading off the second entry completes the proof of Theorem 1.1:

$$\boxed{F_n \;=\; \frac{\varphi^n - \psi^n}{\sqrt{5}} \;=\; \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right).} \qquad \square$$

3. Validity for All n: Proof by Strong Induction

Theorem 1.1 was derived by diagonalization, which implicitly relies on algebraic manipulation. We now give an independent verification that the formula holds for every $n \in \mathbb{N}_0$ via strong induction.

Theorem 3.1. *For all integers $n \geq 0$, $F_n = \dfrac{1}{\sqrt{5}}(\varphi^n - \psi^n)$.*

Proof. Let $P(n)$ denote the statement $F_n = \dfrac{\varphi^n - \psi^n}{\sqrt{5}}$.

Base cases.

- $P(0)$: $\dfrac{\varphi^0 - \psi^0}{\sqrt{5}} = \dfrac{1-1}{\sqrt{5}} = 0 = F_0$. ✓
- $P(1)$: $\dfrac{\varphi^1 - \psi^1}{\sqrt{5}} = \dfrac{\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2}}{\sqrt{5}} = \dfrac{\sqrt{5}}{\sqrt{5}} = 1 = F_1$. ✓

Inductive step. Let $k \geq 1$ and assume $P(j)$ holds for all $j \leq k$. We show $P(k+1)$.

By the Fibonacci recurrence and the inductive hypotheses for $P(k)$ and $P(k-1)$:

$$\begin{aligned} F_{k+1} &= F_k + F_{k-1} \\ &= \frac{1}{\sqrt{5}}\left(\varphi^k - \psi^k\right) + \frac{1}{\sqrt{5}}\left(\varphi^{k-1} - \psi^{k-1}\right) \\ &= \frac{1}{\sqrt{5}}\left[\left(\varphi^k + \varphi^{k-1}\right) - \left(\psi^k + \psi^{k-1}\right)\right] \\ &= \frac{1}{\sqrt{5}}\left[\varphi^k\left(1 + \varphi^{-1}\right) - \psi^k\left(1 + \psi^{-1}\right)\right]. \end{aligned}$$

Substituting $\varphi^{-1} = \dfrac{2}{1+\sqrt{5}}$ and $\psi^{-1} = \dfrac{2}{1-\sqrt{5}}$:

$$F_{k+1} = \frac{1}{\sqrt{5}}\left[\varphi^k\left(1 + \frac{2}{1+\sqrt{5}}\right) - \psi^k\left(1 + \frac{2}{1-\sqrt{5}}\right)\right].$$

Rationalising each term by multiplying numerator and denominator by the conjugate:

$$\begin{aligned}
F_{k+1} &= \frac{1}{\sqrt{5}}\left[\varphi^k\left(1+\frac{2}{1+\sqrt{5}}\cdot\frac{1-\sqrt{5}}{1-\sqrt{5}}\right)-\psi^k\left(1+\frac{2}{1-\sqrt{5}}\cdot\frac{1+\sqrt{5}}{1+\sqrt{5}}\right)\right]\\
&= \frac{1}{\sqrt{5}}\left[\varphi^k\left(1+\frac{2(1-\sqrt{5})}{1-5}\right)-\psi^k\left(1+\frac{2(1+\sqrt{5})}{1-5}\right)\right]\\
&= \frac{1}{\sqrt{5}}\left[\varphi^k\left(1+\frac{2(1-\sqrt{5})}{-4}\right)-\psi^k\left(1+\frac{2(1+\sqrt{5})}{-4}\right)\right]\\
&= \frac{1}{\sqrt{5}}\left[\varphi^k\left(1-\frac{1-\sqrt{5}}{2}\right)-\psi^k\left(1-\frac{1+\sqrt{5}}{2}\right)\right]\\
&= \frac{1}{\sqrt{5}}\left[\varphi^k\left(\frac{2-1+\sqrt{5}}{2}\right)-\psi^k\left(\frac{2-1-\sqrt{5}}{2}\right)\right]\\
&= \frac{1}{\sqrt{5}}\left[\varphi^k\left(\frac{1+\sqrt{5}}{2}\right)-\psi^k\left(\frac{1-\sqrt{5}}{2}\right)\right]\\
&= \frac{1}{\sqrt{5}}\left[\varphi^k\cdot\varphi\ -\ \psi^k\cdot\psi\right]\ =\ \frac{\varphi^{k+1}-\psi^{k+1}}{\sqrt{5}},
\end{aligned}$$

which is precisely $P(k+1)$.

By the principle of strong induction, $P(n)$ holds for all $n \geq 0$. □

4. The Integrality Paradox and Its Resolution

4.1. The Paradox

Binet's formula involves $\sqrt{5}$ in both the numerator and denominator, yet F_n is always a non-negative integer. How can an expression saturated with irrationals produce only integers? For example, at $n = 5$ the formula evaluates to exactly 5.

4.2. Resolution

We show that the irrationals cancel structurally for every n.

Theorem 4.1. *For every $n \geq 1$, φ^n and ψ^n can be written as*

$$\varphi^n = A_n\varphi + B_n, \qquad \psi^n = A_n\psi + B_n,$$

with $A_n, B_n \in \mathbb{N}_0$. Consequently, $F_n = A_n \in \mathbb{N}_0$.

Proof. **Claim.** Since $\varphi^2 = \varphi + 1$, every power of φ can be reduced to a linear combination of φ and 1 with non-negative integer coefficients. The first few steps illustrate the pattern:

$$\begin{aligned}
\varphi^1 &= 1 \cdot \varphi + 0, \\
\varphi^2 &= 1 \cdot \varphi + 1, \\
\varphi^3 &= \varphi^2 \cdot \varphi = (\varphi + 1)\varphi = \varphi^2 + \varphi = 2\varphi + 1, \\
\varphi^4 &= \varphi^3 \cdot \varphi = (2\varphi + 1)\varphi = 2\varphi^2 + \varphi = 2(\varphi + 1) + \varphi = 3\varphi + 2, \\
\varphi^5 &= \varphi^4 \cdot \varphi = (3\varphi + 2)\varphi = 3\varphi^2 + 2\varphi = 3(\varphi + 1) + 2\varphi = 5\varphi + 3.
\end{aligned}$$

The same reduction applies to ψ (since $\psi^2 = \psi + 1$), yielding the *same* coefficients A_n and B_n for both roots.

Cancellation. Substituting into Binet's formula:

$$\begin{aligned}
F_n &= \frac{1}{\sqrt{5}}(\varphi^n - \psi^n) \\
&= \frac{1}{\sqrt{5}}[(A_n\varphi + B_n) - (A_n\psi + B_n)] \\
&= \frac{A_n(\varphi - \psi)}{\sqrt{5}} \\
&= \frac{A_n\sqrt{5}}{\sqrt{5}} = A_n.
\end{aligned}$$

Since $A_n \in \mathbb{N}_0$ for every $n \geq 1$, the formula always produces a non-negative integer. □

Remark 4.2. The coefficients A_n are themselves Fibonacci numbers: $A_n = F_n$ and $B_n = F_{n-1}$. Thus the representation $\varphi^n = F_n\varphi + F_{n-1}$ is a well-known identity in its own right.

5. Applications

5.1. Application 1 — Population Growth (Fibonacci's Rabbits)

Consider the classical model: a pair of rabbits is placed in a field in January. Rabbits take one month to mature; each adult pair produces one young pair per month. The populations of young (y_n) and adult (a_n) pairs in month n satisfy the rules:

1. $y_n = a_{n-1}$ (last month's adults each produce one young pair),
2. $a_n = a_{n-1} + y_{n-1}$ (adults survive and last month's young mature).

Substituting (1) into (2) gives $a_n = a_{n-1} + a_{n-2}$—the Fibonacci recurrence. The table below shows the first ten months.

Month	J	F	M	A	M	J	J	A	S	O
Young pairs	1	0	1	1	2	3	5	8	13	21
Adult pairs	0	1	1	2	3	5	8	13	21	34

The total population in month n is $T_n = a_n + y_n = F_{n+1}$. Binet's formula then gives the population directly:

$$T_n = F_{n+1} = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}\right).$$

5.2. Application 2 — Tiling Combinations

Let W_n denote the number of ways to tile a $1 \times n$ board using 1×1 squares and 1×2 dominoes.

Recurrence. Any valid tiling of a board of length n ends with either

1. a 1×1 square (leaving a $1{\times}(n-1)$ board): W_{n-1} ways, or
2. a 1×2 domino (leaving a $1{\times}(n-2)$ board): W_{n-2} ways.

Since these cases are mutually exclusive and exhaustive:

$$W_n = W_{n-1} + W_{n-2}, \qquad W_1 = 1, \quad W_2 = 2.$$

Comparing with the Fibonacci sequence ($F_2 = 1,\ F_3 = 2$) gives $W_n = F_{n+1}$. By Binet's formula,

$$W_n = F_{n+1} = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}\right).$$

5.3. Application 3 — Planning Study Schedules

Suppose a student is planning a study schedule for n days. On each day, the student can either have a light/rest study day or a heavy study day. However, the student does not want two heavy study days in a row, because this could lead to burnout.

Problem. How many possible study schedules exist for n days if no two heavy study days are consecutive?

Recurrence. Let S_n be the number of valid n-day schedules, where R denotes a rest/light day and H a heavy day. Any valid schedule of length n ends with either

1. a rest/light day R (the first $n-1$ days form any valid schedule): S_{n-1} ways, or
2. a heavy day H, which forces the preceding day to be R (the first $n-2$ days form any valid schedule): S_{n-2} ways.

Since these cases are mutually exclusive and exhaustive:

$$S_n = S_{n-1} + S_{n-2}.$$

This is exactly the Fibonacci recurrence. We now prove $S_n = F_{n+2}$ by induction.

Base cases.

- $n = 1$: the schedules are R and H, so $S_1 = 2 = F_3 = F_{1+2}$. ✓
- $n = 2$: the schedules are RR, RH, HR, so $S_2 = 3 = F_4 = F_{2+2}$. ✓

Inductive step. Let $k \geq 2$ and assume $S_k = F_{k+2}$ and $S_{k-1} = F_{k+1}$. We show $S_{k+1} = F_{(k+1)+2}$.

By the recurrence and the inductive hypotheses:

$$\begin{aligned} S_{k+1} &= S_k + S_{k-1} \\ &= F_{k+2} + F_{k+1}. \end{aligned}$$

By the Fibonacci recurrence relation, $F_{k+2} + F_{k+1} = F_{k+3}$, and since $F_{k+3} = F_{(k+1)+2}$, we conclude $S_{k+1} = F_{(k+1)+2}$.

By the principle of induction, $S_n = F_{n+2}$ holds for all $n \geq 1$. Applying Binet's formula:

$$S_n = F_{n+2} = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^{n+2} - \left(\frac{1-\sqrt{5}}{2} \right)^{n+2} \right).$$

For example, for a 7-day week:

$$S_7 = F_9 = 34.$$

There are exactly 34 valid study schedules for a 7-day week if no two heavy study days are consecutive.

6. Conclusion

This report developed a closed-form expression for the Fibonacci numbers using the tools of linear algebra, and verified its correctness through multiple independent arguments.

In this report we have:

1. Derived Binet's formula (Theorem 1.1) by encoding the Fibonacci recurrence as a linear dynamical systems problem, modeled by the equation $\mathbf{v}_n = M^n \mathbf{v}_0$ and diagonalizing the companion matrix M as $M = PDP^{-1}$, reducing the computation of M^n to the trivial case of a diagonal matrix.

2. Confirmed the formula's universal validity via strong induction (Theorem 3.1), providing an independent verification that does not rely on the algebraic manipulations of the diagonalization argument.

3. Resolved the integrality paradox by showing that every power φ^n and ψ^n admits the representation $A_n\varphi + B_n$ and $A_n\psi + B_n$ with identical integer coefficients $A_n, B_n \in \mathbb{N}_0$, so that the $\sqrt{5}$ terms cancel exactly in Binet's formula (Theorem 4.1).

4. Illustrated the formula through three applications: the classical rabbit population model, the $1\times n$ board-tiling counting problem, and a combinatorial study-schedule planning problem — each of which reduces independently to the Fibonacci recurrence and is therefore solved directly by Binet's formula.

The central lesson of this report is that the Fibonacci sequence, despite its elementary definition, is deeply connected to eigenvalue theory. The fact that a formula involving $\sqrt{5}$ and the golden ratio φ produces an integer for every $n \geq 0$ is not a coincidence — it is a structural consequence of the symmetry between the two eigenvalues φ and ψ, which satisfy the same minimal polynomial and therefore behave identically under integer linear combinations. Matrix diagonalization makes this symmetry explicit and turns what would otherwise be a surprising numerical accident into an inevitable algebraic identity.